

AUTOMATIC FLUENCY EVALUATION OF SPONTANEOUS SPEECH USING DISFLUENCY-BASED FEATURES

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ABSTRACT

This paper describes an automatic fluency evaluation of spontaneous speech. Although we regularly observe a variety of different disfluencies in spontaneous speech, we focus on two types of phenomena, i.e., filled pauses and word fragments. This paper aims to reveal that these two types of disfluencies have effects on speech fluency evaluation differently. To this end, we conduct a series of SVM classification experiments on the Japanese spontaneous speech corpus. The experimental results show that the features derived from word fragments are effective in evaluating disfluent speech especially when combined with prosodic features such as speech rate and pauses/silence, while the features from filled pauses are not effective in evaluating fluency.

Index Terms— Automatic fluency evaluation, disfluency, filled pauses, prosody, spontaneous speech

1. INTRODUCTION

A presentation ability has become important in various scenes. For example, for a lecturer's talk in a university, speaking skill affects the students' understanding [1, 2], and the evaluation on speech fluency of second language learners is important for measuring proficiency in second language learning [3]. Therefore, an automatic evaluation method of how to speak, in particular speech fluency, has been needed. So far, there have been studies on speech fluency evaluation; however most of them have tackled on the second language learning [3, 4, 5].

Among those issues on automatic fluency evaluation, this paper focuses on the issue of disfluency [6, 7, 8, 3, 9, 10]. Although we regularly observe a variety of different disfluencies in spontaneous speech, we focus on two types of phenomena, i.e., filled pauses and word fragments. Previous works on automatic fluency/disfluency evaluation [6, 7, 8, 3, 9, 10] are limited in terms of one or both of those issues of filled pauses and word fragments. Mahesha and Vinod [6] and Kjellgren and Nordstrom [7] are limited in that they did not consider

those two issues. van Dalen et al. [5] focused on disfluency features, however, they did not investigate the effectiveness of each disfluency feature type on fluency evaluation. Deshmukh et al. [8] used some kinds of features on filled pause and word repetition for automatic evaluation of speech fluency but did not consider a word fragment related feature. In addition, the work is insufficient in that it did not make it clear how much disfluency-related features affected fluency evaluation. Overall, the limitation of those previous works is mostly in the lack of considering various phenomena of word fragments.

Considering those limitations of the previous approaches to automatic evaluation of fluency, this paper explores yet another finding on automatic evaluation of fluency/disfluency. This paper especially aims to reveal that the above mentioned two types of disfluencies, namely, filled pauses and word fragments, have effects on speech fluency evaluation differently. To this end, we conduct a series of SVM classification experiments on the Japanese spontaneous speech corpus. The experimental results show that the features derived from word fragments are effective in evaluating disfluent speech especially when combined with prosodic features such as speech rate and pauses/silence, while the features from filled pauses are not effective in evaluating fluency.

2. DATA SET FOR EVALUATION

Corpus of Spontaneous Japanese (CSJ) [11, 12] is a large-scale database that includes spontaneous speeches (lecture etc.) in Japanese. It contains speech signal and transcription of about 7 million words along with various annotations like POS and phonetic labels. Table 1 shows an example of the transcription of CSJ, which is a portion of the transcript of an example of the “disfluent” class to be described later in this section. In CSJ, as shown in Table 1, recorded speech is transcribed in two different ways: orthographic and phonetic transcriptions¹. In orthographic transcription,

¹Each line of a transcript of CSJ corresponds to a Japanese *bunsetsu*, consisting of one or more content words such as nouns and verbs, followed by zero or more functional words such as particles, auxiliary verbs, and suffixes.

Table 1. An example of the CSJ transcription (an example of the “disfluent” class).

ID	start ~ end (sec.)	orthographic transcription		phonetic transcription			⟨filled -pause⟩ ⟨word -fragments⟩ tags
		in Kanji/Kana letters	English translation	in Kana letters	in syllable representation	# morae	
—	268.338 ~ 268.869	⟨pause⟩					
0127	268.869	(F あのー)	uh	(F アノー)	a no o	3	⟨filled -pause⟩ —
	~ 271.138	何か	well	ナニカ	na ni ka	3	
—	271.138 ~ 271.791	⟨pause⟩					
0128	271.791	(F ま)	so	(F マ)	ma	1	⟨filled -pause⟩ — — — ⟨word -fragments⟩
	~ 273.823	郵便配達の人	a mail delivery person	ユービンハイタツノヒトワ	yu u bi N ha i ta tsu no hi to wa	9	
		あれでしょうけど	might be probably	アレデショーケド	a re de sho o ke do	3	
		(D ん)	what?	(D ン)	N	7	
						1	
—	273.823 ~ 274.268	⟨pause⟩					

Table 2. Statistics of the data set.

class	# speech data (# speakers)	# files (each around 10 sec. duration)	mean opinion score of fluency-disfluency rating (MOS, 7-ranks score averaged over 10 annotators)
fluent	54	494	5.0 ~ 6.2
neutral	109	1,422	3.2 ~ 4.9
disfluent	38	253	1.9 ~ 3.1
total	201	2,169	1.9 ~ 6.2

speech is transcribed using Kanji (Chinese logograph) and Kana (Japanese syllabary) just like ordinary Japanese text. In phonetic transcription, on the other hand, its transcription is written exclusively in Kana letters so that the phonetic details of the utterance being transcribed can be traced. More detailed transcription as well as the description of tags used for the annotation are found on the Web site of CSJ². Various tags were embedded in these transcriptions to mark phenomena specific to spontaneous speech like filled pauses, word fragment, reduced articulation, mispronunciation, etc.

In this paper, among the transcription of CSJ, we utilize

²In a transcript, several lines of bunsetsus are then delimited by a line representing a *pause*, where, in CSJ, duration of silence over 200 ms is considered as a *pause*.

²https://pj.ninjal.ac.jp/corpus_center/csj/misc/preliminary/5.html

Table 3. Features.

(a) Prosodic features.

feature name	definition	code
speech rate	average number of morae per sec.	SpR
# pauses per mora	ratio of total number of pauses to total number of morae	Ps/Mr
silence rate	ratio of contiguous silence to duration	SiIR

(b) Disfluency-based features.

feature name	definition	code
# filled pauses per mora	ratio of total number of filled pauses to total number of morae	FP /Mr
# word fragments per mora	ratio of total number of word fragments to total number of morae	WF /Mr
mora length of filled pauses per mora	ratio of total mora length of filled pauses to total number of morae	MrFP /Mr
mora length of word fragments per mora	ratio of total mora length of word fragments to total number of morae	MrWF /Mr

the following information in the evaluation: the duration as well as pause or silence information, the mora length of utterances measured in terms of character length of the phonetic transcription, and filled pauses as well as word fragments information (as shown in Table 1) as the most important disfluency-related information.

The sources of speech data of CSJ consist of 89 academic presentation speeches and 112 simulated public speeches. In CSJ, to each of those 201 speech data, rated impressions of public speaking such as “liking”, “skillfulness”, “speech rate”, “activity”, and “formality” are annotated [12]. Among those rated impressions of public speaking, this paper utilizes that of 7-ranks rating of fluency-disfluency out of the “skill-

Table 4. Feature values and mean opinion scores of fluency-disfluency rating: averages of fluent/neutral/disfluent classes.

class	prosodic features				disfluency-based features								mean opinion score (MOS, 7-ranks score averaged over 10 anno-tators)
	SpR	pauses		SilR	filled pauses				word fragments				
		total # (Ps)	Ps /Mr		total # (FP)	FP /Mr	total mora length (MrFP)	Mr FP /Mr	total # (WF)	WF /Mr	total mora length (MrWF)	Mr WF /Mr	
fluent	7.94	3.3	0.038	0.125	15.0	0.0294	27.9	0.0547	2.1	0.0041	3.1	0.0062	5.3
neutral	6.84	3.8	0.052	0.187	14.8	0.0334	28.8	0.0653	2.9	0.0067	4.5	0.0105	4.1
disfluent	5.53	4.4	0.075	0.285	15.9	0.0443	31.2	0.0883	4.1	0.0115	6.7	0.0187	2.8

fulness” ratings. 10 annotators rated each speech data according to 7-ranks rating of fluency-disfluency, where we utilize their average over 10 annotators. Then, as shown in Table 2, we classify the total 201 (speech and its transcription) data into the following three classes: “fluent” whose average ratings range from 5.0 to 6.2, “neutral” whose average ratings range from 3.2 to 4.9, and “disfluent” whose average ratings range from 1.9 to 3.1. Finally, we divide each speech and its transcription data into its constituent files, each of which is of around 10 seconds duration, obtaining 2,169 files in total (as shown in Table 2)³. In the experimental evaluation of this paper, we pursue the following two way splits of those three classes, i.e., (i) fluent (494 files) vs. neutral and disfluent (1,675 files), and (ii) fluent and neutral (1,916 files) vs. disfluent (253 files). Then, we evaluate our proposed method in two binary classification tasks of fluent speech detection and disfluent speech detection.

3. FEATURES OF SVM

Table 3 lists features of SVM employed in this paper. Table 3(a) lists three prosodic features, namely, the speech rate, number of pauses per mora⁴, and the ratio of the contiguous silence to the duration of the speech. Table 3(b), on the other hand, lists the disfluency-based features employed in this paper, which are obtained from the transcription of the CSJ. As we showed in Table 1, the number and the mora length of filled pauses as well as word fragments are available in the phonetic transcription of CSJ, which we utilize as the disfluency-based features in this paper. Table 4 shows average values of those 7 features and the mean opinion scores in each of the fluent / neutral / disfluent classes. In the table,

³More specifically, within each of the total 201 speech data, 7-ranks rating of fluency-disfluency is annotated to its constituent portion of around 50 seconds or more duration, but not to the whole speech duration. Thus, sometimes it can happen that one of the 201 speech data has both a “fluent” rated portion and a “neutral” rated portion, or both a “disfluent” rated portion and a “neutral” rated portion. In those cases, we remove those “neutral” rated portions and only keep “fluent” rated or “disfluent” rated portions.

⁴The number of morae is measured as the Kana length of the phonetic transcription of the CSJ corpus.

features other than the number of filled pauses per mora and the mora length of filled pauses per mora are in **bold faced**, because those bold faced features mostly contribute to achieving optimal recall/f-measure in this paper.

4. EXPERIMENT

We evaluate our framework of automatic fluency evaluation by SVM through 5-fold cross validation. In this paper, we apply the SVM toolkit of scikit-learn [13] package to the task of automatic classification of fluency evaluation. In each data split of the 5-fold cross validation, we perform the grid-search⁵ of the kernel functions (rbf, linear, and 2nd degree polynomial) and the hyper parameters C and γ , and evaluate them against the development set⁶.

Table 5 shows results when running SVM with each single feature out of the employed 7 features, where the kernel functions and the hyper parameters are optimized through the grid search by maximizing recall or f-measure. Both of Table 5(a) and Table 5(b) rank those individual features in descending order of the optimized recall or f-measure. One of the most important findings of those tables is that, roughly speaking, prosodic features and word fragment features outperform filled pauses features in both binary classification tasks. Meanwhile, it is also obvious that those higher ranked features perform better when optimizing *f-measure* of detecting *disfluent* speech than in that of detecting *fluent* speech. It is also interesting to note that word fragments features perform much better in detecting *disfluent* speech than in detecting *fluent* speech. It can be clearly stated that these findings quite coincide with our intuition.

Table 6 further examines the experimental results of feature combinations selected from the list of the exhaustive combinations of all the 7 features. For each of the feature

⁵In both of the binary classifications of fluent speech detection and disfluent speech detection, we used the three types of target optimization functions: i.e., recall, precision, and f-measure of the target class to be detected.

⁶Due to the shortage of the data size in the experiment of this paper, at the moment, we evaluated the proposed method against the development data set. In the final version of this paper, we perform training / development / test data split and evaluate the proposed method against the held-out test data set.

Table 5. Results of running SVM with a single feature by optimization through grid search (individual features are ranked in descending order of optimized recall/f-measure (%)).

(a) Binary classification of fluent vs. neutral-disfluent classes.

rank	optimizing recall	optimizing f-measure
1	Ps/Mr 85.3	SiIR 39.5
2	SpR 83.9	SpR 38.7
3	SiIR 79.5	Ps/Mr 37.3
4	WF/Mr 77.8	MrWF/Mr 37.0
5	MrFP/Mr 76.9	MrFP/Mr 34.7
6	MrWF/Mr 76.0	WF/Mr 34.5
7	FP/Mr 63.5	FP/Mr 30.6

(b) Binary classification of fluent-neutral vs. disfluent classes.

rank	optimizing recall	optimizing f-measure
1	MrWF/Mr 85.1	WF/Mr 55.6
2	SpR 83.3	SpR 49.7
3	SiIR 79.9	MrWF/Mr 49.4
4	WF/Mr 72.0	SiIR 46.9
5	Ps/Mr 70.4	Ps/Mr 46.3
6	FP/Mr 59.1	FP/Mr 39.9
7	MrFP/Mr 58.1	MrFP/Mr 34.0

combinations, again, the kernel functions and the hyper parameters are optimized through the grid search by maximizing recall or f-measure. Then, we compare feature combinations which satisfy one of the following four requirements: (i) achieving the highest recall/f-measure, (ii) achieving the highest recall/f-measure among those without any of the four disfluency-based features, (iii) achieving the highest recall/f-measure among those without any of the two filled pauses features, (iv) achieving the highest recall/f-measure among those without any of the two word fragments features. Overall, it is obvious that the proposed approach achieved higher recall/f-measure in detecting *disfluent* speech than in detecting *fluent* speech. Also, as for the comparison of feature combinations satisfying one of those four requirements (i) ~ (iv), it is quite impressive to note that feature combinations of word fragments features and prosodic features without any of the two filled pauses features achieved (almost or sometimes exactly) the highest recall/f-measure in both tasks. This result indicates that, in the task of detecting fluent/disfluent speech, both of the word fragments and the prosodic features are inevitable, while it is not necessary to incorporate the filled pauses features in those tasks. Furthermore, it is also clear in the result of Table 6(b) that the differences of the recall/f-measure in Table 6(b) are much larger compared to those differences in Table 6(a). This supports the claim that the contribution of those word fragments and prosodic features is much more in detecting *disfluent* speech than in detecting *fluent* speech.

5. CONCLUDING REMARK

This paper showed that the features derived from word fragments were effective in evaluating disfluent speech, while the

Table 6. Experimental results of feature combination (%). (a) Binary classification of fluent vs. neutral-disfluent classes (recall/precision/f-measure of detecting fluent class).

features		(macro ave.)
baseline (minority)		100 / 22.8 / 37.1
optimizing recall	SpR + Ps/Mr + WF/Mr (highest recall = highest recall w/o filled pauses)	89.7 / 24.1 / 37.9
	SpR + Ps/Mr (highest recall w/o disfluency-based features)	88.4 / 24.0 / 37.6
	SpR + Ps/Mr + SiIR + MrFP/Mr (highest recall w/o word fragments)	84.8 / 24.9 / 38.2
optimizing f-measure	SpR + Ps/Mr + SiIR + FP/Mr + MrWF/Mr (highest f-measure)	61.4 / 35.2 / 44.1
	SpR + SiIR + WF/Mr + MrWF/Mr (highest f-measure w/o filled pauses)	71.5 / 32.0 / 43.6
	SiIR + MrFP/Mr (highest f-measure w/o word fragments)	71.3 / 31.0 / 42.7
	Ps/Mr + SiIR (highest f-measure w/o disfluency-based features)	82.3 / 26.6 / 40.0

(b) Binary classification of fluent-neutral vs. disfluent classes (recall/precision/f-measure of detecting disfluent class).

features		(macro ave.)
baseline (minority)		100 / 11.7 / 20.9
optimizing recall	6 features (w/o MrFP/Mr) (highest recall)	96.9 / 27.8 / 43.0
	SpR + SiIR + WF/Mr (highest recall w/o filled pauses)	96.6 / 27.3 / 41.9
	SpR + Ps/Mr + SiIR (highest recall w/o disfluency-based features)	88.3 / 27.1 / 41.3
	SiIR + FP/Mr (highest recall w/o word fragments)	87.7 / 27.0 / 41.0
optimizing f-measure	SpR + WF/Mr (highest f-measure = highest f-measure w/o filled pauses)	72.3 / 53.7 / 61.5
	SpR + SiIR + FP/Mr (highest f-measure w/o word fragments)	80.1 / 40.2 / 53.3
	SpR + Ps/Mr (highest f-measure w/o disfluency-based features)	69.9 / 42.4 / 52.3

features from filled pauses were not. In our preliminary evaluation, we incorporated acoustic features within the framework of LSTM-based time series fluency/disfluency classification, while the time series acoustic features did not contribute to the task. Our definite future work includes employing a number of automatic techniques of detecting various disfluencies such as filled pauses and word fragments within speech and text [14, 15, 16, 17, 18, 19, 20, 21, 22]. Here, acoustic features should help detecting disfluencies, and hence, are expected to contribute to our framework based on disfluency-based features. Further extension to automatic evaluation of impressions of public speaking [23] other than fluency/disfluency should be another direction of future work.

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6. REFERENCES

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